SR series/TS2000/TS2100
Industrial Robot

INSTRUCTION MANUAL

WAFER TRANSFER ROBOT

SR–404HC,
SR–405HC–L1550–R/L
[with Traverse Axis (Actuator) BB70]

MAINTENANCE

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before installing and operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Rev.1
April 1, 2020

Shibaura Machine

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PREFACE

This manual describes how to maintain the robot and controller.
Maintenance and inspection are essential to keep the robot performance for a long time, to prevent failures and to assure safety. Be sure to read through this manual and set up a maintenance program before actually starting the robot.

This manual is composed of the following sections.

Section 1: Maintenance Items
This section describes the maintenance schedule of the robot and controller and items required for inspection and maintenance.

Section 2: Maintenance of the Robot
This section refers to the robot structure and items required for inspection and maintenance of the robot.

Section 3: Maintenance of Controller
This section refers to the robot controller structure and items required for inspection and maintenance of the controller.

CAUTIONS ON SAFETY

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will imminently lead to fatalities or serious injuries."
 **WARNING**	This means that "incorrect handling may lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

! CAUTION

- Be sure to read through the Safety Manual provided separately before starting actual work to assure safety of the work covering from the robot installation to operation.

[Maintenance and inspection]

Be sure to observe the following items to use the robot safely.

! DANGER	
 Prohibited	• DO NOT incinerate, disassemble or charge the batteries. Otherwise, they may rupture.
 Mandatory	• Be sure to turn off the main power switch of the controller before starting inspection or maintenance. • Batteries should be disposed of according to the user's in-house regulations.

! CAUTION	
 Disassembly prohibited	• The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.
 Mandatory	• Always use SHIBAURA MACHINE's designated spare parts when replacing the parts. • Maintenance and inspection should be performed regularly. Otherwise, the system may malfunction or accidents will be caused.

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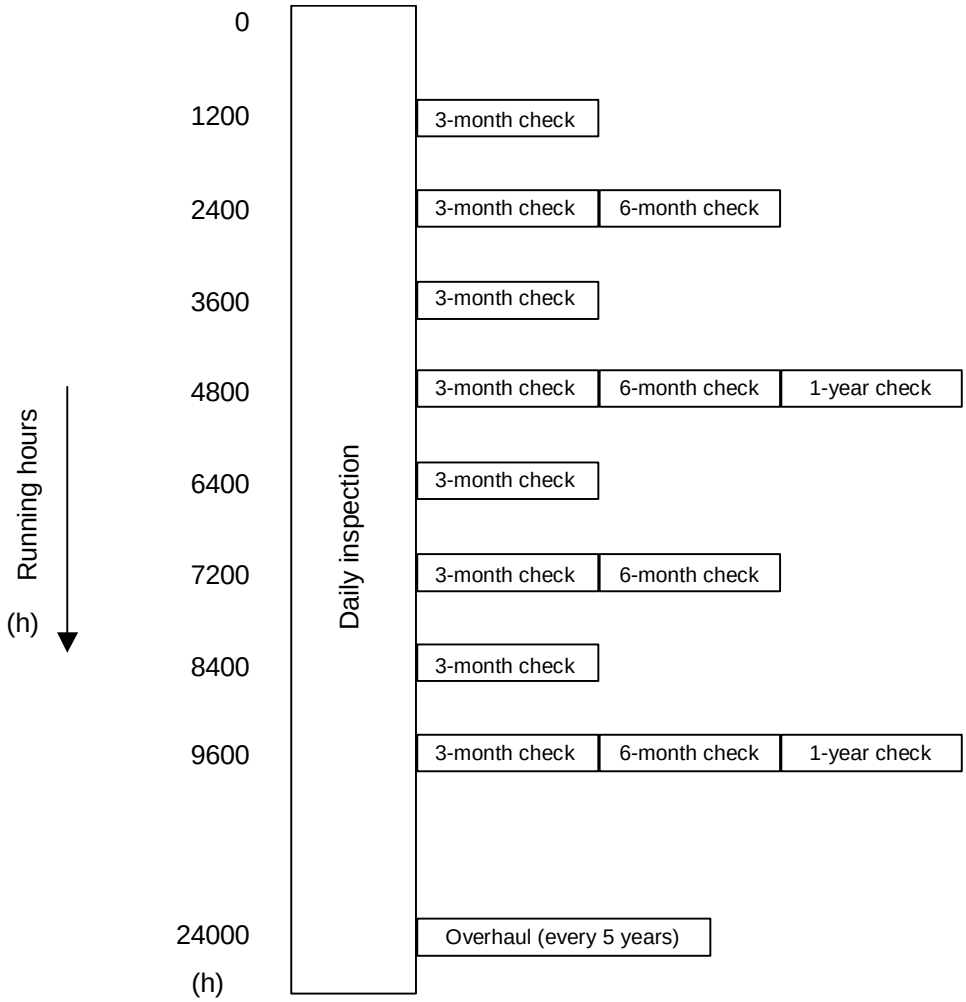
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Section 1 Maintenance Items

1.1 Maintenance Schedule

Maintenance comes in the two (2) types; daily inspection, and regular inspection and maintenance. For the regular inspection and maintenance, inspection items are added every 1,200 running hours.



Inspection schedule (yardstick)

- When the system is operated for 16 hours per day:
  16 hours × 25 days × 3 months = 1,200 hours
- When the system is operated for 24 hours per day:
  24 hours × 25 days × 3 months = 1,800 hours

1.2 Items for Maintenance and Inspection

This section describes the items for maintenance and inspection. For the execution procedures, see the relevant paragraph listed in the table below.

a) Daily inspection

No.	Item	Inspection procedures	Relevant Para.
1	Robot installation bolts	Make sure that the robot base installation bolts are tightened completely.	2.3.1
2	Vibration and noise	Make sure that each axis can move smoothly.	2.3.1
3	Change in positioning accuracy	Make sure that the robot can stop at a predetermined position accurately.	
4	Controller vent holes	(1) Make sure that no obstacle stands in front of the vent holes. (2) Make sure that no contaminant is left in the vent hole.	3.3.1
5	Function of safety devices	Make sure that each safety device connected to the emergency stop pushbutton switch and external operation input signal can function properly.	3.3.1

b) Quarterly (every three (3)-month) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Lubricated state of ball screw and linear guide	Make sure that the ball screw and linear guide of each axis are greased completely and that no rust is developed on them.	2.3.2

c) Semi-annual (every six (6)-month) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Wiring	Check for the state of the flexible cables and connectors.	2.3.3
2	Each axis drive system	Move each axis by hand and make sure that it moves smoothly.	2.3.1
3	Timing belts	(1) Make sure that each timing belt is not worn off or cracked. (2) Check each timing belt for tension.	2.3.4

d) Annual (every year) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Greasing	Apply the grease to each axis ball screw and linear guide.	2.3.2

e) Battery replacement

No.	Item	Inspection procedures	Relevant Para.
1	Replacement of position detector batteries (Replace the batteries every three (3) years.)	Replace the position detector backup batteries. If the running time is long, battery consumption is small. If the power OFF time is long, the consumption is large. If the battery voltage has dropped, the battery alarm lamp flickers. Replace the batteries immediately then.	2.3.5

f) Overhaul (every five (5) years)

No.	Item	Inspection procedures	Relevant Para.
1	Inspection of consumable parts	Inspect the ball screws, linear guides, flexible cables, reduction gears, bearings and motors and replace them when necessary.	
2	Battery replacement (Replace the battery every five (5) years.)	Replace the RAM (controller) backup battery. If the running time is long, battery consumption is small. If the power OFF time is long, the consumption is large. If the battery voltage has dropped, the battery alarm lamp illuminates. Replace the battery immediately then. Alarm message: "MAIN Battery alarm"	3.3.3
3	Replacement of switching power supply units (Replace the units every five (5) years.)	As the switching power supply unit uses an aluminum electrolytic condenser, its life is limited. Replace each switching power supply unit (DC5 V, DC24 V) for preventive maintenance. The life of the switching power supply unit differs with load current, running hours and ambient temperature. If the load current is large, the running time is long and the ambient temperature is high, the life becomes short.	3.3.4

1.3 Maintenance Tools

We recommend to use the following tools for maintaining the robot and controller.

- Phillips screwdrivers
- Hexagonal wrench set: M3 ~ M10
- Assembly wrench set
- Water pump pliers
- Nippers
- Grease guns

1.4 Maintenance Contract and Repair

1.4.1 Maintenance Contract

With the exception of greasing and lubrication, we recommend that the user conclude with us an after-sale service contract on the inspection items for maintenance and inspection to be done semi-annually and after.

1.4.2 Repair

If a fault has occurred or if repair is necessary, turn off the controller power and contact the SHIBAURA MACHINE Service Department. At this time, advise us of the details of the fault and the following information stated on the robot and controller.

- Robot model
- Manufacture number
- Date of manufacture
- Operating time

1.4.3 Modification

This robot and controller MUST NOT be modified without the consent from SHIBAURA MACHINE.



CAUTION

- The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.

Section 2 Maintenance of the Robot

2.1 Cautions on Maintenance and Inspection

When performing inspection or maintenance of the robot, strictly observe the following precautions to protect yourself and coworkers.

! DANGER

- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.

2.2 Layout of Robot Components and Drive Mechanism

The layout of the robot mechanical components is shown in Fig. 2.1.

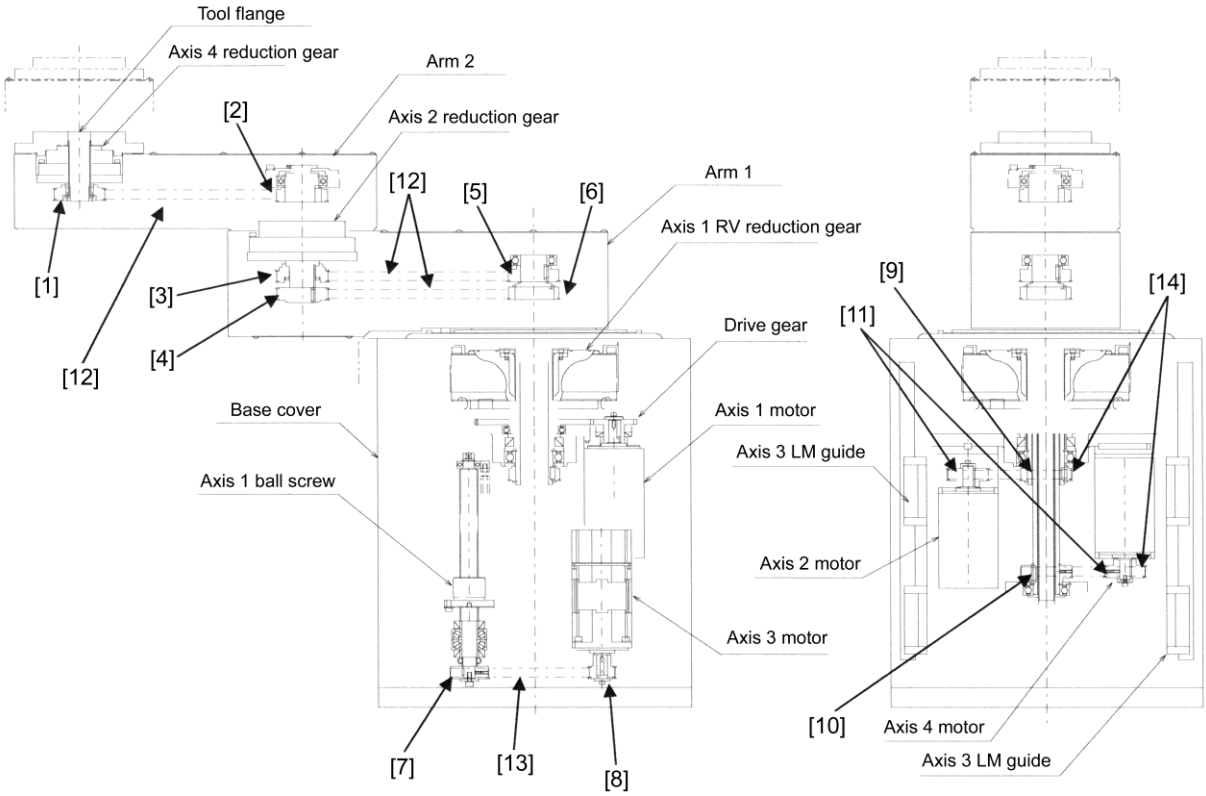
a) Axis 1
The output of axis 1 AC servo motor is input to the RV reduction gear (made by Teijin Seiki) via the spur gear. The output of the RV reduction gear directly controls the arm 1 motion.

b) Axis 2
The output of axis 2 AC servo motor is input to the harmonic drive via the timing belt. The output of the harmonic drive directly controls the arm 2 motion.

c) Axis 3
The output of axis 3 AC servo motor is input to rotate the ball screw via the timing belt to move the vertical axis up and down. The axis 3 simultaneously moves up and down the three (3) motions of axis 1 arm, axis 2 arm and axis 4 hand.

d) Axis 4
The output of axis 4 AC servo motor is input to the harmonic drive via the timing belt. The output of the harmonic drive directly controls the axis 4 hand rotation.

e) Axis 5 (only for the SR–405HC–L1550)
The output of axis 5 AC servo motor is input to rotate the ball screw. The output of the ball screw directly controls the slider motion.



Part numbers [1] to [11] are the timing pulleys and part numbers [12] to [14] the timing belts.

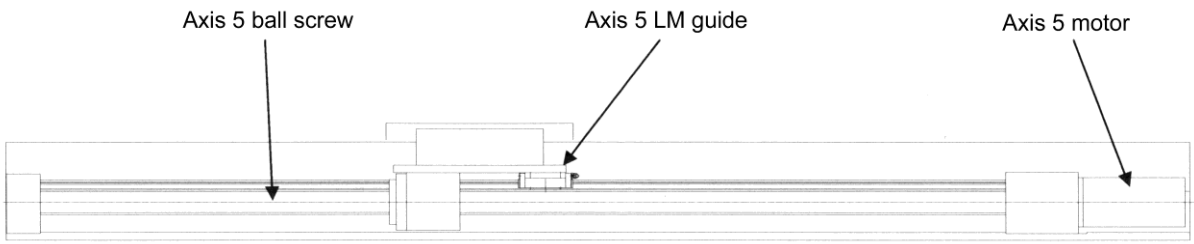


Fig. 2.1 Layout of robot mechanical components

2.3 Robot Maintenance Procedures

2.3.1 Check for Vibration and Abnormal Noise

1) Approach the robot and its peripheral equipment outside the predetermined operating range and make sure that no abnormal noise of operation is heard. Otherwise, locate the source of the abnormal noise.
When necessary, turn off the power of the robot and its peripheral equipment and manually move the part suspected to be the trouble source to confirm that the abnormal noise is heard from it.
If the abnormal noise is heard from the ball screw and linear guide, contaminant has possibly entered them, or they have run out of grease. For the former cause, remove the contaminant and make sure that other parts are not damaged. For the latter cause, apply grease.

2) If chattering is caused in the cover, etc., make sure that all screws are tightened completely and that relevant parts are not deformed. Otherwise, completely tighten the screws or repair the parts as necessary.

3) Make sure that the screws of each arm joint are tightened completely. Otherwise, tighten the screws completely.

4) Possible causes other than the above-mentioned are the wrong method of use, controller-related cause, etc. In either case, adjustment is not easy. Consult us at the after-sale service agent in your territory.

! DANGER

- When nearing the robot in the servo-free status, be sure to carry out the work by two (2) persons. One person should carry out the work while the other is monitoring the work outside the hazardous zone. The latter person should watch the work and be ready to press the emergency stop switch if any abnormality occurs.

2.3.2 Check for Lubrication

For the RV reduction gear of axis 1, use the Moly-White grease (made by Teijin Seiki).
For the axis 3 (vertical axis), use the FOMBLIN grease OT–20 (made by Solvay Solexis).
For the axis 5 (traverse axis), use the LG2 (made by NSK).

For the reduction gear in the arm, the following grease is used. The reduction gear is of a component type, and need not be greased again, however. If an abnormal noise or backlash is found before overhaul, consult us at the after-sale service agent in your territory.

Grease for arm reduction gear: 4B No.2 (Harmonic drive)

1) Check of ball screw and linear guide for lubrication state
   Make sure that the ball screw and linear guide of each axis are greased sufficiently and that no rust has developed on them. If they have run out of grease, fill it.

! CAUTION

- NEVER mistake the kind of grease. Otherwise, the robot performance will deteriorate or the robot will malfunction.
- After you have touched the ball screw, ball screw spline or linear guide by hand, be sure to fill the grease. Otherwise, rust will develop on it.

2) Greasing

   i) Dismounting of base cover.
      When you have removed the two-split base cover, you can see the linear guide there.

   ii) Greasing of linear guide:
      Apply the grease to the rails.

   iii) Greasing of ball screw:
      Fill the grease to the roller surface of the ball screw shaft.

iv) Mounting of base cover:
Set the two-split base cover.

The greasing volume is as follows:

- Ball screw ⋯ Approx. 5 g
- Linear guide ⋯ Approx. 5 g (per slider)

v) Greasing of traverse axis
See the ROIBOT actuator instruction manual provided separately.

2.3.3 Check for Wiring

1) Make sure that the cable surface in the base cover is not damaged and in a good state.
Open the arm 1 and arm 2 covers and make sure that each cable surface is not damaged, or stretched or slackened due to unusual force exerted on it.

2) If the cable surface is damaged, identify the cause and repair. If the cable interior is also damaged, replace the cable with a new one.

2.3.4 Check for Timing Belt Status

Remove the cover and visually check to see that the belt is not worn off or cracked. Also check the timing belt for tension and adjust as necessary.

2.3.5 Replacement of Position Detector (or Encoder) Batteries

! CAUTION

- Batteries should be disposed of according to the user's in-house regulations.
- NEVER drop the batteries into fire. NEVER short-circuit, charge, disassemble or heat them. Otherwise, liquid leakage or rupture may be caused.

To back up the data of the position detector attached to each motor, batteries (lithium dry batteries) are used. Replace the batteries every three (3) years.
If the controller has detected the battery voltage drop alarm, replace the batteries at an early stage.
For the battery replacement, observe the following procedures according to the battery power ON time.

! CAUTION

- When the battery voltage has dropped, a battery alarm will generate. If the batteries are replaced just after generation of the battery alarm, the battery voltage returns to normal with the battery alarm reset automatically. Unless the batteries are changed just after generation of the battery alarm, however, the battery voltage will drop further and a battery error will occur. Under this condition, position data detected by the encoder is not reliable. As a result, a position detection error occurs and the robot enters an emergency stop state so that the servo system cannot be turned on. If the power is turned off in this condition, the position data is lost.

  To avoid the above, be sure to replace the batteries with new ones at the time of triennial (every three (3)-year) inspection.

2.3.6 Replacing Batteries

DANGER

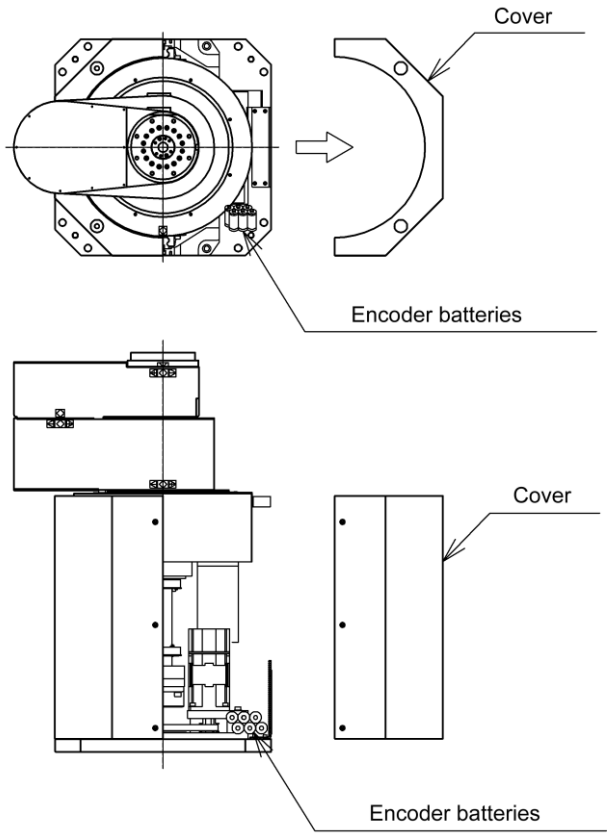
- Only after assuring the safe work, replace the batteries with new ones while the power is turned on and an emergency stop is effected on the robot.

1) Turn the power off.

2) Remove the cover.

3) Replace the batteries with new ones within five (5) minutes. If the batteries are to be replaced while the power is turned off, start replacing them after one (1) minute has passed and finish replacing within five (5) minutes, because power failure operating current flows for several tens seconds at power OFF.

4) Set the new batteries, then the cover.



2.3.7 Check for Resin Belt

Make sure that the resin belt of the traverse axis is not loosened or damaged. To replace the belt, see the ROIBOT actuator instruction manual provided separately.

Section 3 Maintenance of Controller

3.1 Cautions on Maintenance and Inspection

When performing maintenance and inspection of the controller, follow the items given below so that the work can be carried out safely.

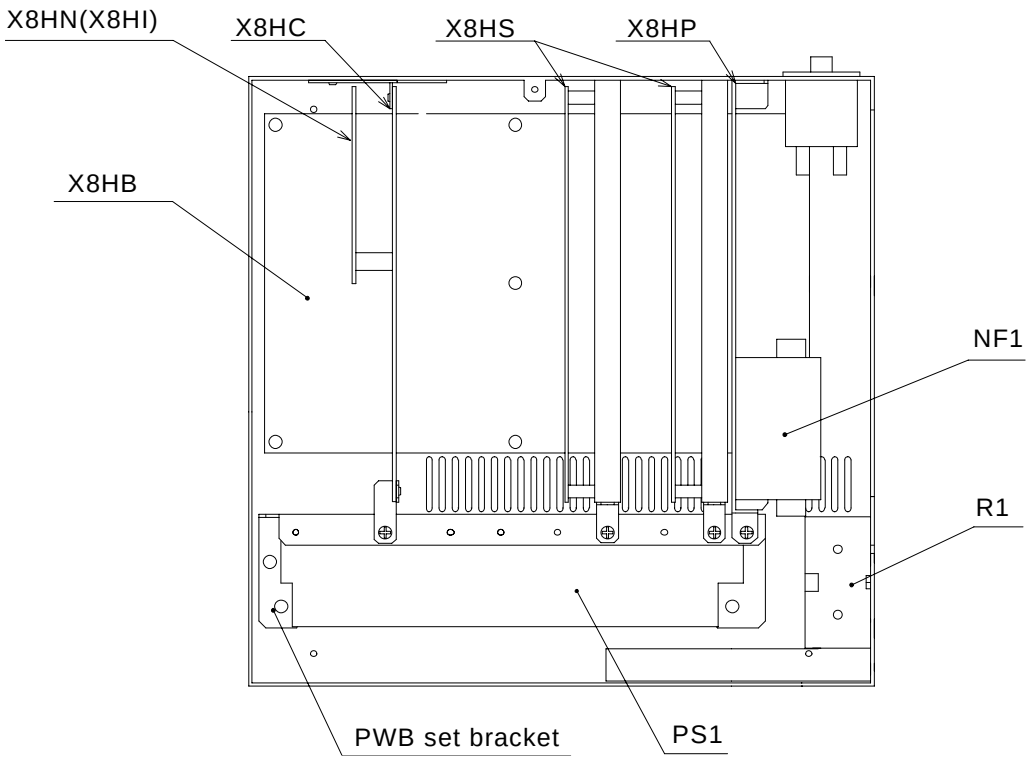
! CAUTION

- Before removing the controller cover for maintenance or inspection, be sure to turn off the main power switch of the controller.
  Also, turn off the power breaker when connecting or disconnecting the power cable.
- DO NOT touch the servo driver for at least two (2) minutes after the main power switch is turned off. The large-capacity capacitor in the servo driver may be charged with voltage, and touching the servo driver could lead to electric shocks.
- DO NOT disconnect the battery connector at other than battery replacement. Otherwise, files saved in the memory may be lost.

! DANGER

Be sure to turn off the main power switch. If the main power switch remains ON, the servo power printed board, servo printed board and switching power supply are charged, and touching them could lead to electric shocks.

3.2 Layout of TS2000 Controller Parts

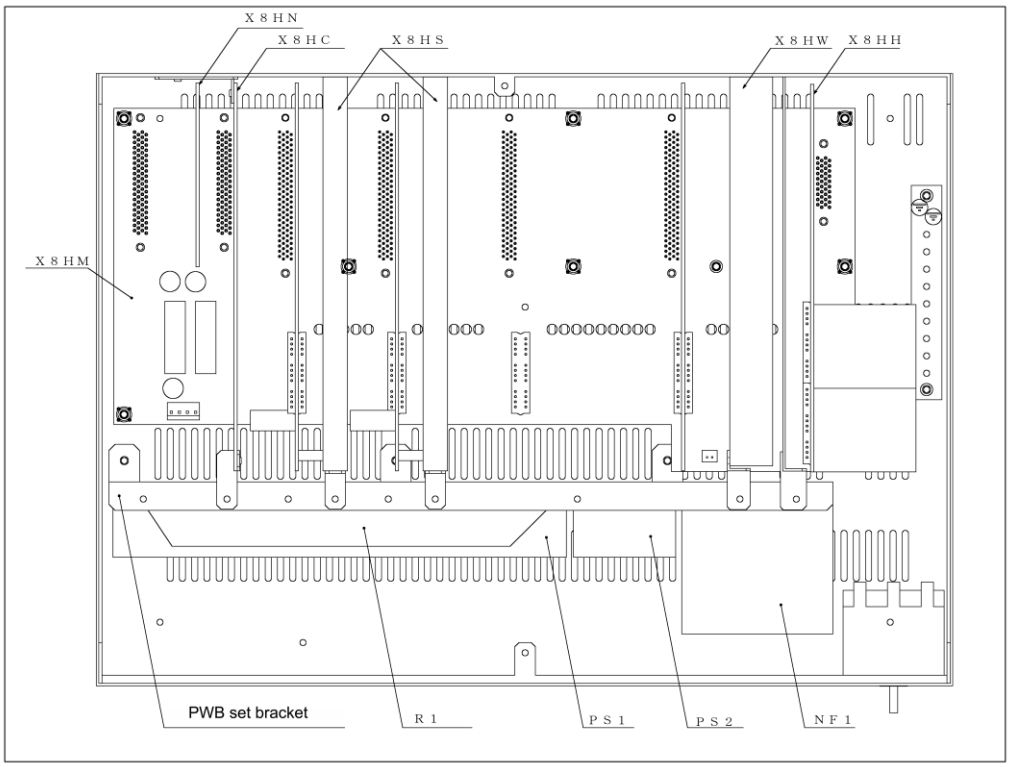


Upper side view

Fig. 3.1 Layout of TS2000 controller parts

Part name	Descriptions
PS1	P5 V, P24 V switching power supply
PWB set bracket	PS1 set bracket
X8HP	Servo power printed board
X8HC	Main control printed board
X8HN (X8HI)	I/O output printed board
X8HS	Servo printed board
X8HB	Mother board
R1	Regenerative resistor
NF1	Noise filter

3.3 Layout of TS2100 Controller Parts



Upper side view

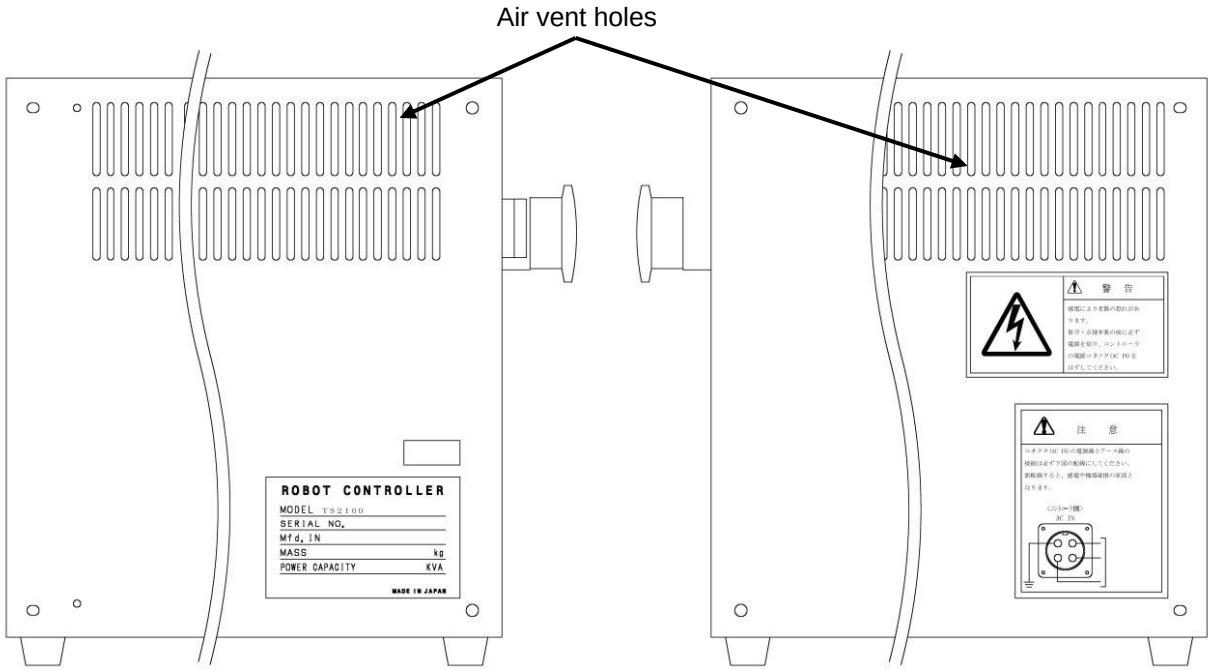
Fig. 3.2 Layout of TS2100 controller parts

Part name	Descriptions
PS1	P5 V, P24 V switching power supply
PS2	P24 V switching power supply
PWB set bracket	PS1, etc. set bracket
X8HH	Servo power printed board
X8HC	Main control printed board
X8HN (X8HI)	I/O output printed board
X8HS, X8HW	Servo printed board
X8HM	Mother board
R1	Regenerative resistor
NF1	Noise filter

3.4 Maintenance Procedures

3.4.1 Check for Controller Air Vent Holes

If the air vent holes are blocked, the controller may overheat and malfunction. To avoid this, perform check on the air vent holes to make sure that air is flowing freely through them.



Left side view Right side view

Fig. 3.3 Controller side views

a) Make sure that no obstacle stands in front of the air vent holes.
If there is any obstacles, move them away from the vent holes so that the air flow is not blocked.

b) Make sure that no contaminant is left on the air vent holes. If any contaminant is left on the vent hole, remove it.

c) If the controller is placed sideways, make sure that the air vent holes on the bottom surface of the controller are more than 11 mm away from the floor surface. Always keep the space of 11 mm or over. The length of rubber shoe is approximately 11 mm.

3.4.2 Check of Safety Devices for Function

Make sure that the emergency stop pushbutton switches equipped on the control panel and teach pendant work properly. Also make sure that the safety devices controlled by the external operation input signals work correctly.

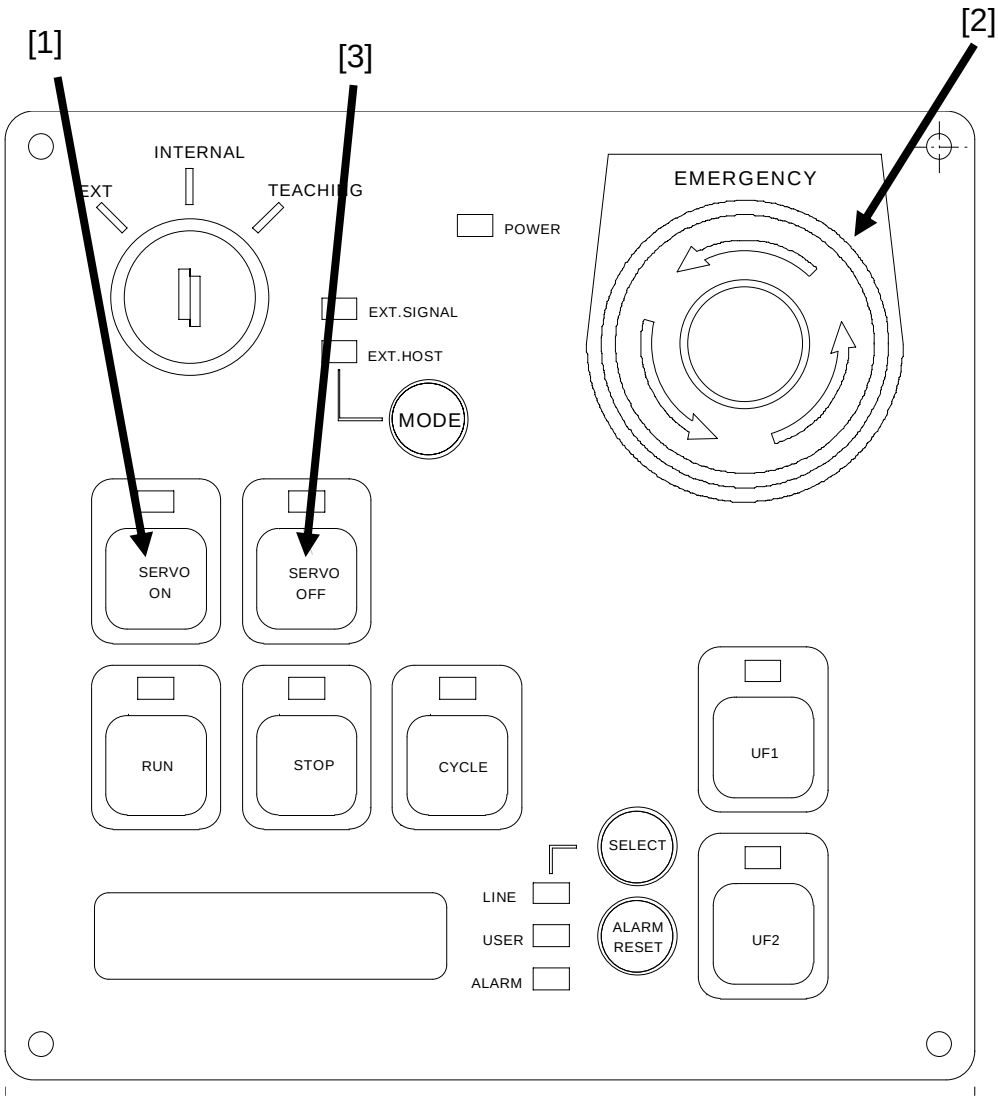


Fig. 3.4 Layout of control panel switches

a) Make sure that the EMERGENCY (emergency stop) switch [2] provided on the control panel functions properly.

1) Press the SERVO ON switch [1] provided on the control panel to turn the servo on and make sure that the SERVO ON LED is illuminated.

2) Press the EMERGENCY switch [2] and make sure that the servo is turned off. The SERVO ON LED goes out and the SERVO OFF LED illuminates. At this time, make sure that the EMERGENCY switch remains depressed.

3) Turn the EMERGENCY switch clockwise and make sure that the same switch turns off.

b) Make sure that the EMERGENCY STOP switch [6] provided on the teach pendant functions properly.

1) Press the SERVO ON switch [4] on the teach pendant to turn on the servo power, and make sure that the SERVO ON LED [5] is illuminated.

* The servo ON is not possible in the TEST RUN mode.

2) Press the EMERGENCY STOP switch [6] on the teach pendant and make sure that the servo is turned off. The SERVO ON LED goes out. At this time, make sure that the EMERGENCY STOP switch [6] remains depressed.

3) Turn the EMERGENCY STOP switch in the clockwise (CW) direction and make sure that the same switch turns off.

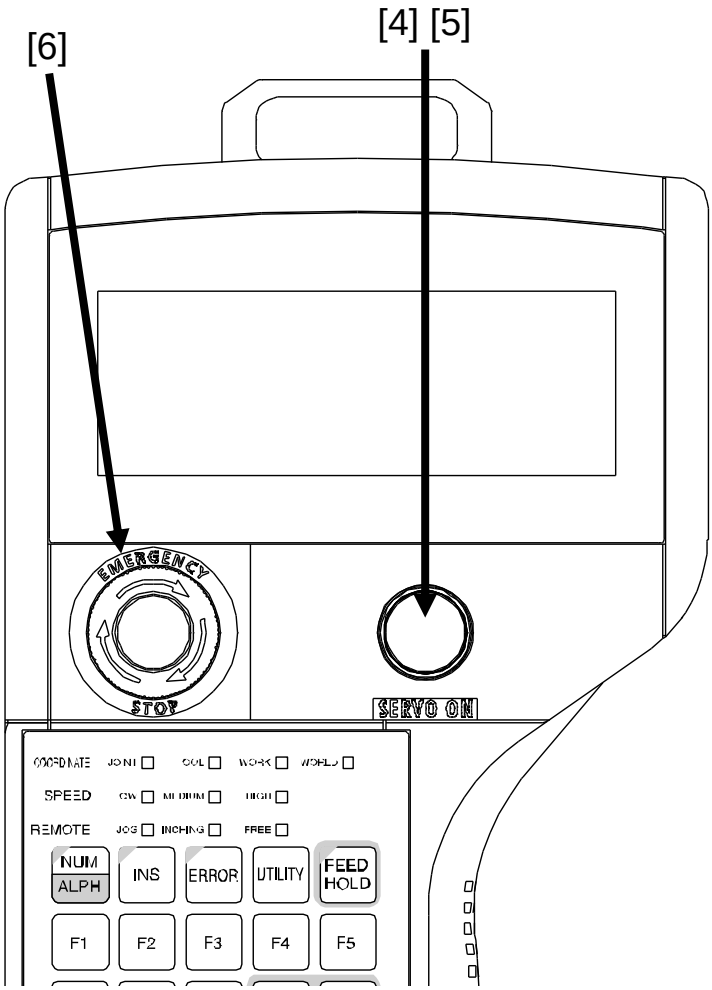


Fig. 3.5 EMERGENCY STOP switch provided on teach pendant

c) Make sure that the safety devices controlled by the external operation input signals work properly.

1) Press the SERVO ON switch [1] equipped on the control panel to turn on the servo power. Then turn on (or trip) the safety devices connected with external operation input signal "emergency stop" (such as external emergency stop switch, photoelectric type sensing safety device and foot switch) to make sure that the servo power is turned off.

2) Set ON the safety devices connected with external operation input signal "low speed command", and make sure that each signal functions properly.

3.4.3 Battery Replacement

The memory equipped on the X8HC printed board of the robot controller is backed up by a lithium battery to save the data. Replace the battery every five (5) years. The lithium battery will turn its life when used for a predetermined time. If it is used, neglecting the life, the battery voltage will drop to below the voltage required for keeping the memory data, resulting in data being lost and faults caused by leakage of the battery liquid.

If the following alarm has occurred, replace the battery with a new one.

MAIN Battery alarm (alarm code 1–145)

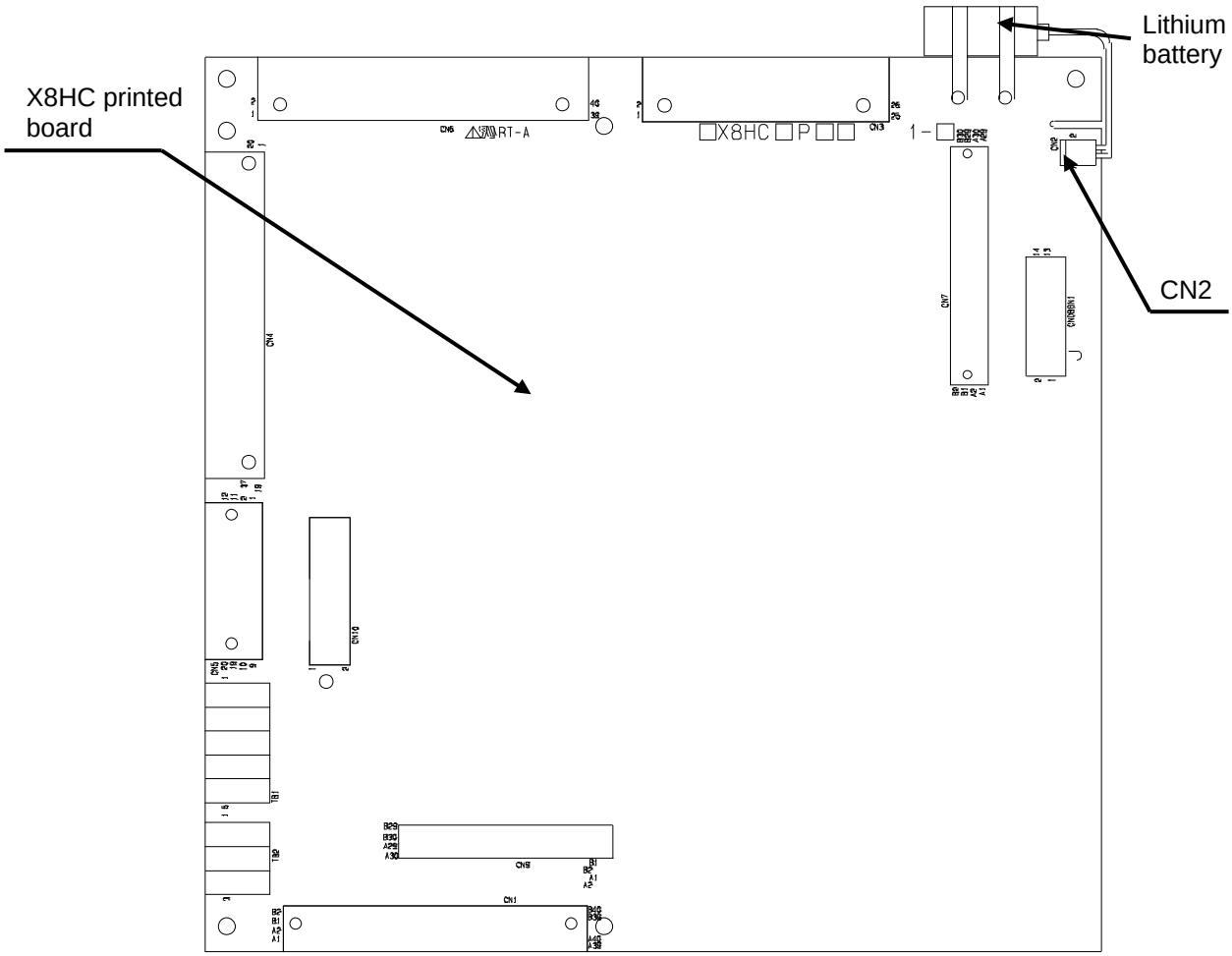


Fig. 3.6 Location of lithium battery

Table 3.1 Type of battery

Type of battery	Manufacturer	Battery service life
ER3V-C	TOSHIBA LIFESTYLE PRODUCTS & SERVICES Corp.	10 years (at room temperature)

Note: The battery life is ten (10) years at the room temperature. As it differs with the external environment such as operating temperature and humidity, we recommend to replace the battery every five (5) years.

Only the battery shown in Table 3.1 should be used. As this is an exclusive battery, contact SHIBAURA MACHINE at order entry.

[Battery replacement procedures]

1) When the battery is to be kept removed for more than one (1) minute, save all programs stored in the memory into the personal computer beforehand. In the worst case, all memory data are lost.

2) Set OFF the main power switch.

3) Remove the cover from the controller.

4) Remove the battery connector from CN2 of the X8HC printed board, then disconnect the battery from the Tie-wrap securing the battery.
(For the battery position, see Fig. 3.6.)

5) Set a new battery within one (1) minute.
If the battery is kept removed for more than one (1) minute, all memory data are lost.
The lithium battery capacity decreases by natural discharge even if not used. Use a lithium battery which is within three (3) years (kept at the room temperature) from purchasing.

6) Set the cover to the controller.

7) If the battery has been removed for more than one (1) minute, set ON the main power switch and load the programs from the personal computer where the TSPC (option) is already installed.

If the controller has detected the battery voltage drop alarm even before the five (5)-year replacement period, replace the battery immediately.

! CAUTION

- Batteries should be disposed of according to the user's in-house regulations.
- NEVER drop the batteries into fire. NEVER short-circuit, charge, disassemble or heat them. Otherwise, liquid leakage or rupture may be caused.

3.4.4 Replacement of Switching Power Supply Units

The life of the switching power supply units (DC5 V, DC24 V) used in the robot controller differs with the operating conditions. These units use an aluminum electrolytic capacitor, and if the load current is large, the running time is long and the ambient temperature is high, the life will reduce.
As the life varies largely with the user's operating conditions (5 ~ 10 years), replace the power supply units every five (5) years for preventive maintenance.

[Replacement procedures]

1) Turn off the main power switch.

2) Turn off the power breaker.

3) Remove the cover from the controller.

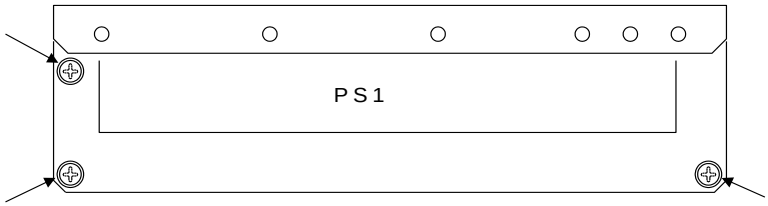
4) Remove all connectors in the control panel, which are linked with the X8HC, X8HS (2 pcs.), X8HP, X8HH and X8HW printed boards. (For the position of each printed board, see Fig. 3.1 and Fig. 3.2. The X8HH and X8HW printed boards are used when the TS2100 robot controller is used.)

5) Remove the X8HC, X8HN (X8HI), X8HS (2 pcs.), X8HP, X8HH and X8HW printed boards from the body. (The X8HH and X8HW printed boards are used when the TS2100 robot controller is used.)

6) Disconnect the cables connecting the parts on the PWB set bracket.

7) Remove the screws (3 places) which secure the body and PWB set bracket. (For the position of the PWB set bracket, see Fig. 3.1 and Fig. 3.2.)

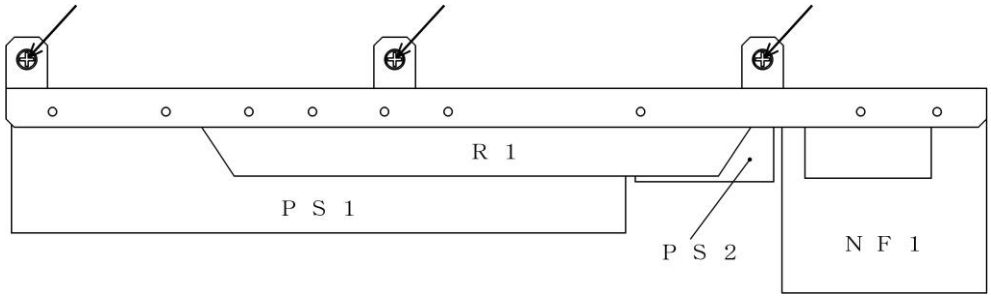
- PWB set bracket of TS2000



Upper side view

Fig. 3.7 PWB set bracket of TS2000 (upper side view)

- PWB set bracket of TS2100



Upper side view

Fig. 3.8 PWB set bracket of TS2100 (upper side view)

8) Remove the PWB set bracket from the body.

9) Disconnect respective power supply units from the PWB set bracket and set the new power supply units.

- PWB set bracket of TS2000

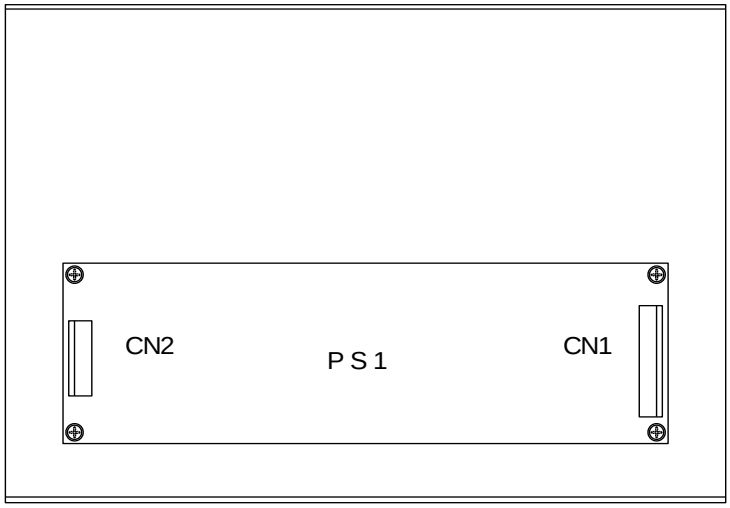


Fig. 3.9 PWB set bracket of TS2000

- PWB set bracket of TS2100

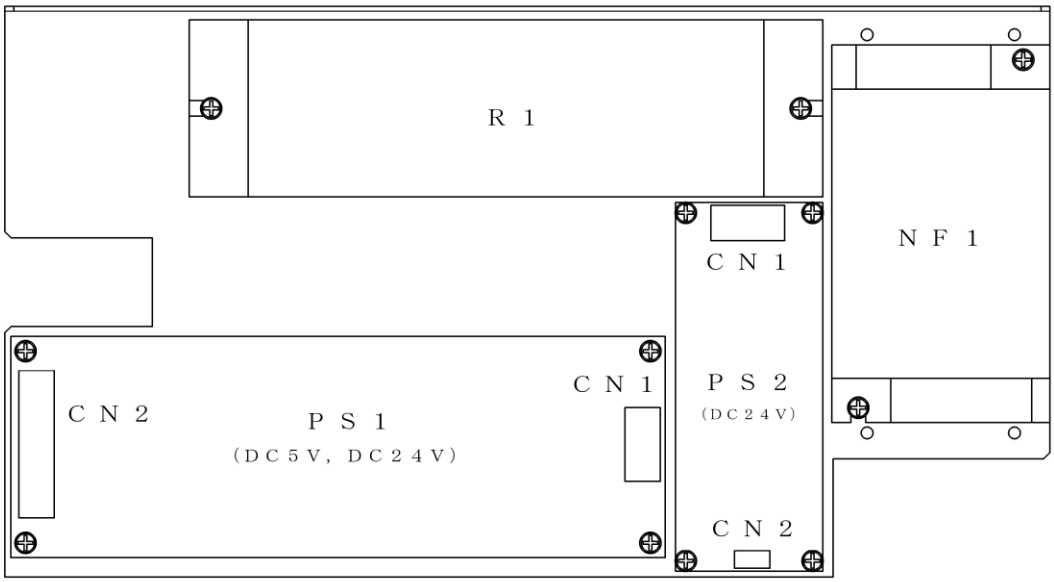


Fig. 3.10 PWB set bracket of TS2100

10) Return the PWB set bracket where the new power supply units are mounted to the controller, then connect the connectors.

11) Screw the PWB set bracket into the body. (Three (3) places; See Fig. 3.7 or Fig. 3.8.)

12) Mount the dismantled printed boards on the body.

13) Connect the disconnected connectors with respective printed boards.

14) Attach the cover to the controller.

3.4.5 Replacement of Fuse (X8HN, X8HI Printed Boards)

If the current exceeding the specified current has run through the I/O unit, the fuse of the X8HN (X8HI) printed board is blown out. If the alarm saying "I/O FUSE Broken (8–273)" is displayed, replace the fuse with a new one.
If the fuse of the output module has been blown out, examine and identify an fault circuit, remedy the cause, then replace the fuse.

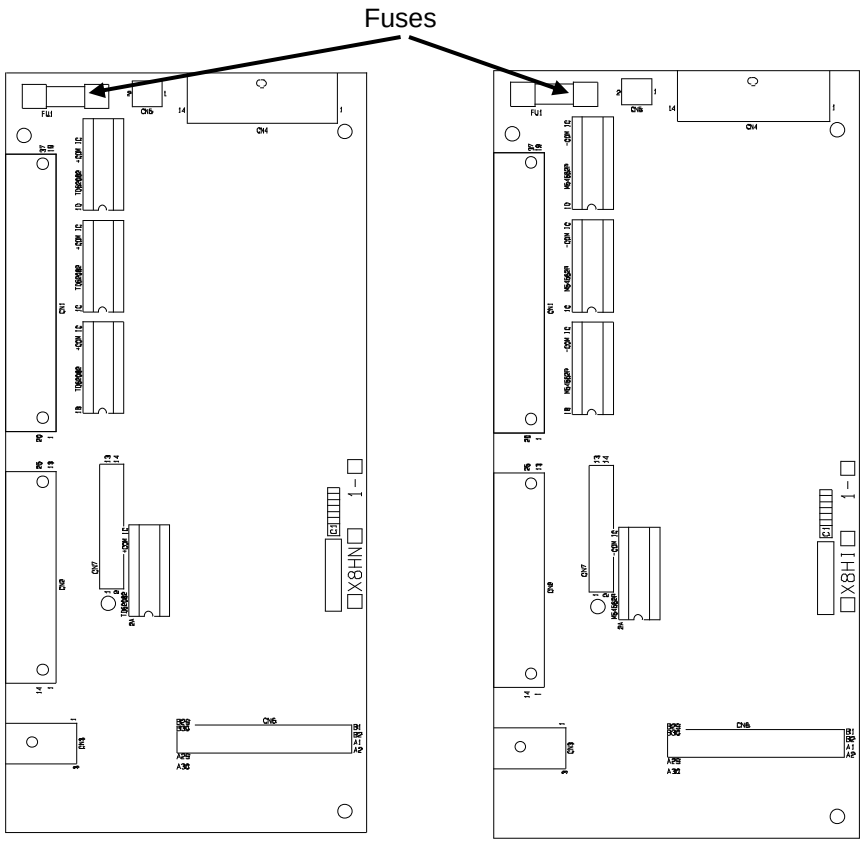


Table 3.2 Type of fuse

Type of fuse	Manufacturer
51NM030H	PICO

[Replacement procedures]

1) Turn off the controller main power.

2) Remove the cover attached to the fuse unit as shown above.

3) Remove the fuse and set a new fuse.

4) Mount the detached fuse cover.

5) Turn on the controller main power and make sure that no error will occur.

3.4.6 Replacement of Output ICs (X8HN, X8HI Printed Boards)

If the current exceeding the specified current has run through the output unit, the outputs may not turn on normally.
When this happens, the ICs of the X8HN (X8HI) printed board have possibly been damaged. Replace the ICs.
Before the replacement, examine and identify an fault circuit, then replace the ICs.

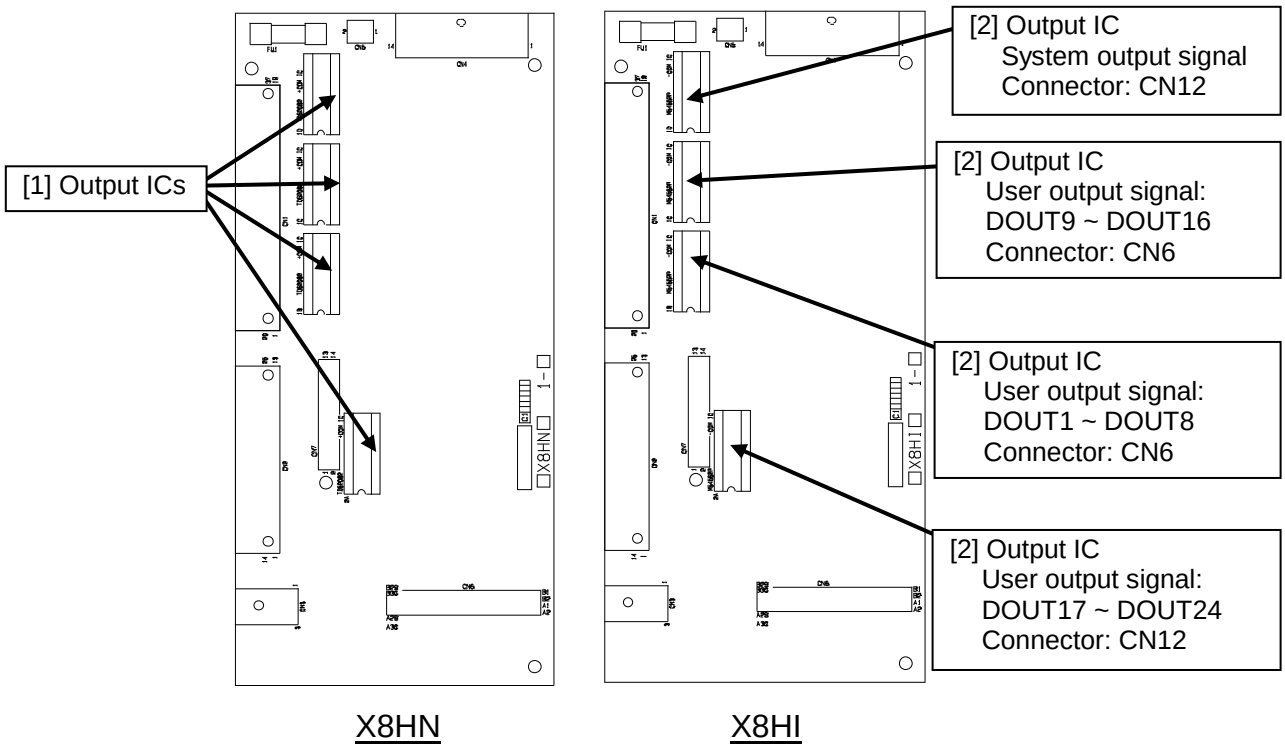


Table 3.3 Type of IC

	Name of board	Type of IC	Manufacturer
[1]	X8HN	TD62082AP	TOSHIBA Corp.
[2]	X8HI	M54562P	Mitsubishi Electric

[Replacement procedures]

1) Turn off the controller main power.

2) Remove the cover from the body.

3) Remove all connectors in the control panel, which are linked with the X8HC printed board.

4) Remove the X8HC and X8HN (X8HI) printed boards from the body.

5) Remove the ICs shown above from respective sockets.

6) Attach new ICs to the sockets.

7) Mount the X8HC and X8HN (X8HI) printed boards on the body.

8) Link the disconnected connectors with the X8HC printed board, and set the main cover.

9) Turn on the controller main power.

Appendix Table 1 List of Mechanical Replacement Parts for Maintenance (Parts for SR–404, SR–405HC–L1550)

Part name	Type	Maker	Q'ty	Remarks
AC servo motor	TS4513N3124E200 (600W)	SHIBAURA MACHINE	1	Axis 1
	TS4512N3120E200 (400W)		1	Axis 2
	TS4512N3120E200 (400W)		1	Axis 4
	TS4512N8120E200 (400WB)		1	Axis 3
Reduction gear	RV–50C–32, 54–A–T	SHIBAURA MACHINE	1	Axis 1
	SHF–32–80–2UH–SP	SHIBAURA MACHINE	1	Axis 2
	SHF–25–80–2UH–SP		1	Axis 4
Timing belt	513–3GT–12	SHIBAURA MACHINE	1	Axis 3
	624–3GT–12 (in the arm)		1	Axis 2
	408–3GT–12 (in the base)		1 each	Axes 2, 4
	624–3GT–12 (in the arm)		2	Axis 4
Timing pulley	S727312 (drive side)	SHIBAURA MACHINE	1	Axis 3
	S727313 (driven side)		1	Axis 3
	S727314 (drive side in the base)		1 each	Axes 2, 4
	S727315 (driven side in the base)		1	Axis 4
	S727316 (driven side in the base)		1	Axis 2
	S727303 (drive side in arm 1)		1	Axis 2
	S720384 (driven side in arm 1)		1	Axis 2
	S720385 (drive side in arm 2)		1	Axis 4
	S720386 (driven side in arm 2)		1	Axis 4
	S700122 (drive side in arm 1)		1	Axis 4
	S720382 (driven side in arm 1)		1	Axis 4
Support unit	FK–20	SHIBAURA MACHINE	1	Axis 3
Ball screw	M060979	SHIBAURA MACHINE	1	Axis 3
LM guide	M067824	SHIBAURA MACHINE	1	Axis 3
Lithium battery	KK1613G01	SHIBAURA MACHINE	2	
Grease	Moly-White grease	Teijin Seiki		
	FOMBLIN grease OT–20	Solvay Solexis		
	LG2 (for traverse axis)	NSK		

Appendix Table 2 List of SR–404HC Harness Replacement Parts for Maintenance

Part name	Type	Maker	Q'ty	Remarks
Harness * For details, see Appendix Drawing 2.	KK7628G01	SHIBAURA MACHINE	1	
	M170225		1	
	M170226		1	
	KL1702G01 (Cable length between robot and controller: 5 m)		1	CN2
	KL1703G01 (Cable length between robot and controller: 5 m)		1	CN3
	KL1704G01 (Cable length between robot and controller: 5 m)		1	CN4

Appendix Table 3 List of SR–405HC–L1550 Harness Replacement Parts for Maintenance

Part name	Type	Maker	Q'ty	Remarks
Harness * For details, see Appendix Drawing 2.	KK7628G01	SHIBAURA MACHINE	1	
	M170225		1	
	M170226		1	
	300N4401 (in the traverse axis)		1	
	300N4501 (in the traverse axis)		1	
	300N4601 (in the traverse axis)		1	
	300N4701 (in the traverse axis)		1	
	300N3901 (Cable length between robot and controller: 5 m)		1	CN2
	300N4001 (Cable length between robot and controller: 5 m)		1	CN3
	300N4101 (Cable length between robot and controller: 5 m)		1	CN4
	300N4201 (Cable length between robot and controller: 5 m)		1	CN11

! CAUTION

- With the exception of the grease, all parts listed above are custom-made to SHIBAURA MACHINE specifications. Contact SHIBAURA MACHINE at order entry.
- NEVER replace the parts by the user.

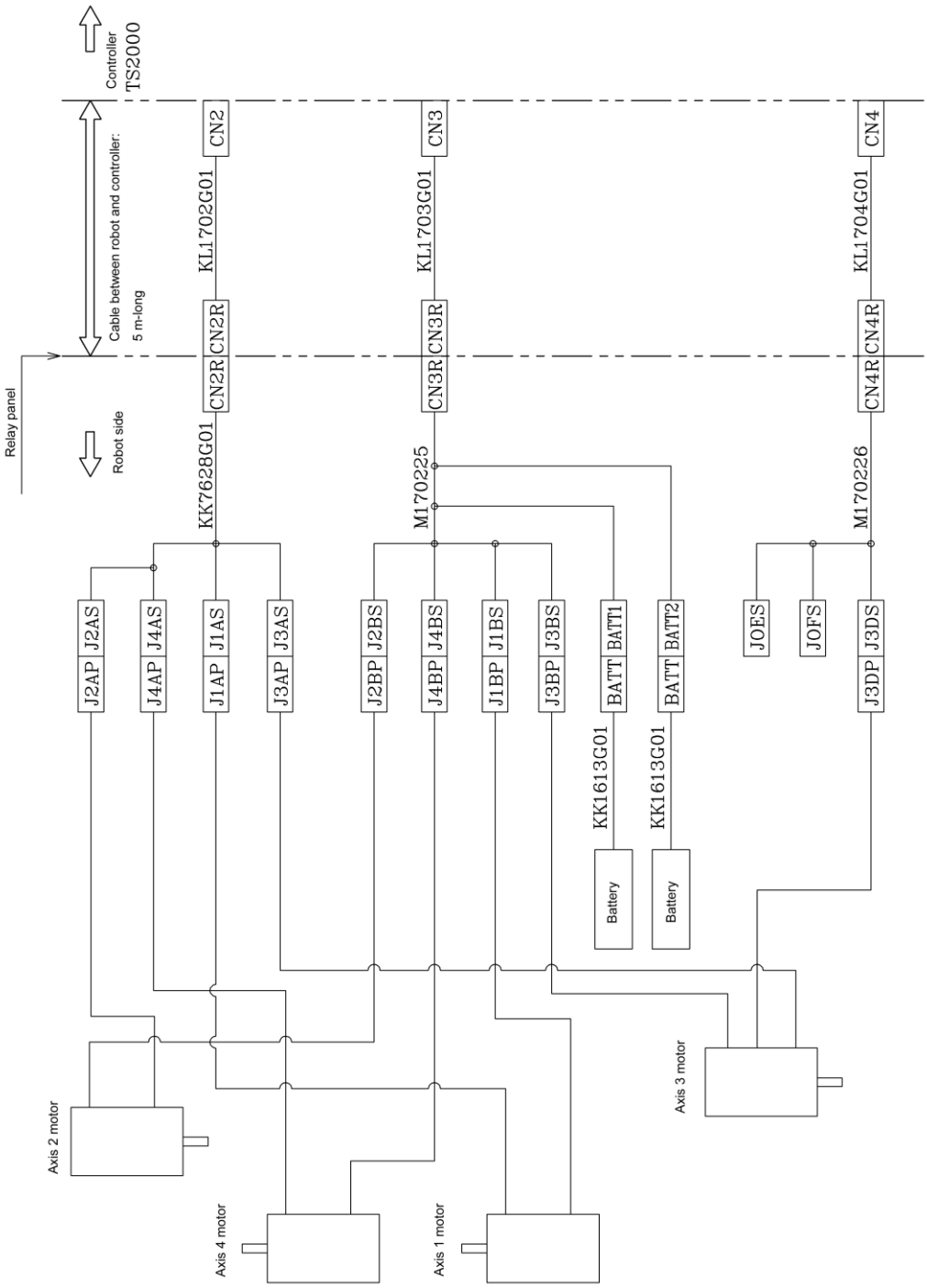
Appendix Table 4 List of TS2000 Controller Replacement Parts for Maintenance (for SR–404HC)

Part name	Type	Maker	Q'ty	Remarks
PS1 (Switching power supply)	LEB100F–0524	Cosel	1	
Main power switch	IR–11–AB–10A–1B	Nippon Thermo	1	
Lithium battery	ER3V-C	TOSHIBA LIFESTYLE PRODUCTS & SERVICES Corp.	1	
Fuse (X8HC printed board)	51NM030H	PICO	1	
Printed circuit board	X8HC (main control)	SHIBAURA MACHINE	1	
	X8HN, X8HI (I/O printed board)		1	
	X8HP (servo power supply)		1	
	X8HB (mother board)		1	
	X8HJ (connector)		1	
Operation panel unit	X8HO	SHIBAURA MACHINE	1	
Teach pendant	TP1000	SHIBAURA MACHINE	1	With 5m-long cable
System disk	TS2000SYS	SHIBAURA MACHINE	1	

Appendix Table 5 List of TS2100 Controller Replacement Parts for Maintenance (for SR–405HC–L1550)

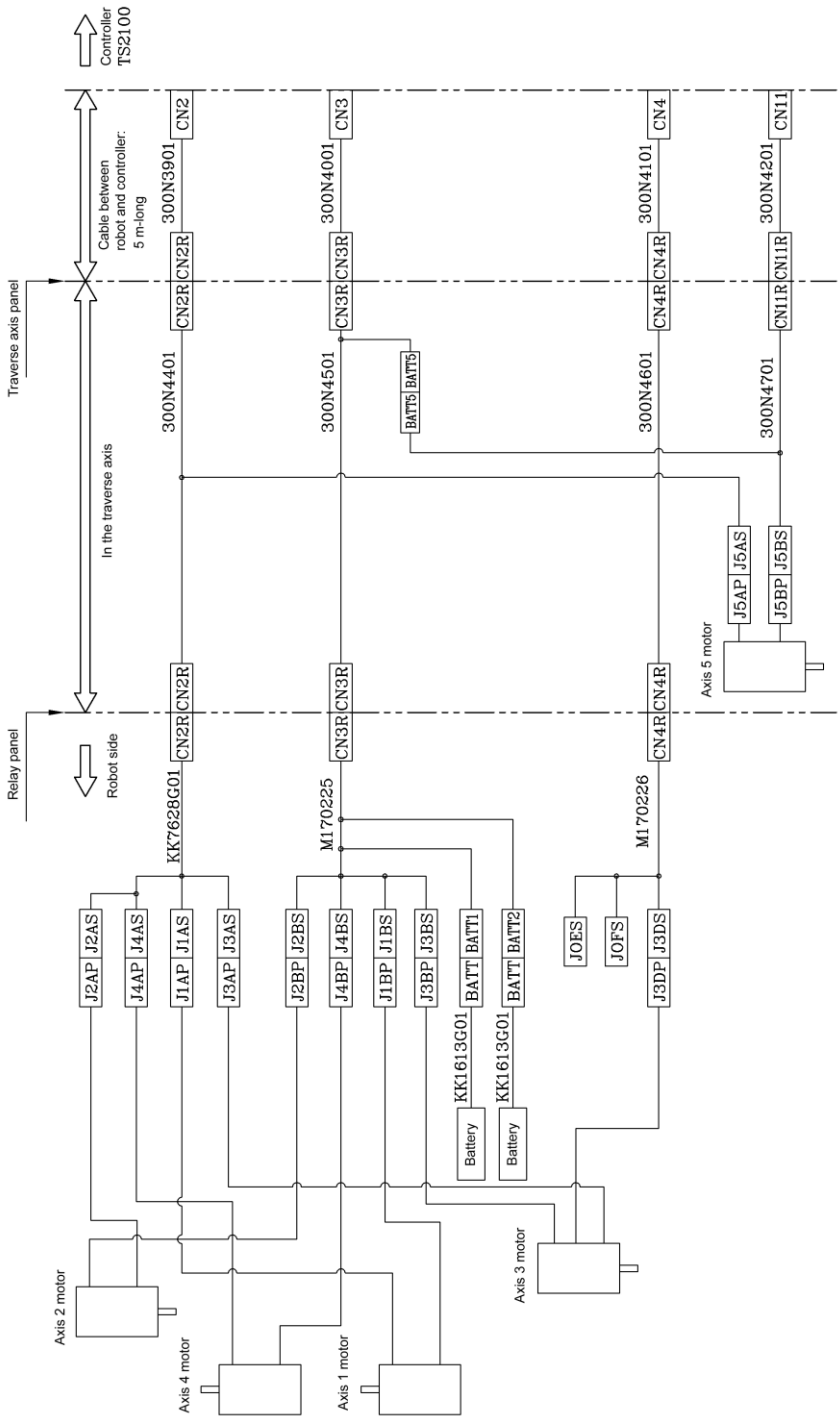
Part name	Type	Maker	Q'ty	Remarks
PS1 (Switching power supply)	LEB100F–0524	Cosel	1	
PS2 (Switching power supply)	LDA15F–24	Cosel	1	
Main power switch	ICP25–L–111–108–10	Nippon Thermo	1	
Lithium battery	ER3V-C	TOSHIBA LIFESTYLE PRODUCTS & SERVICES Corp.	1	
Fuse (X8HC printed board)	51NM030H	PICO	1	
Printed circuit board	X8HC (main control)	SHIBAURA MACHINE	1	
	X8HN, X8HI (I/O printed board)		1	
	X8HH (servo power supply)		1	
	X8HM (mother board)		1	
	X8HJ (connector)		1	
Operation panel unit	X8HO	SHIBAURA MACHINE	1	
Teach pendant	TP1000	SHIBAURA MACHINE	1	With 5m-long cable
System disk	TS2000SYS	SHIBAURA MACHINE	1	

Appendix Drawing 1 Component Connection Diagram (TS2000/SR–404HC)



Component connection diagram

Appendix Drawing 2 Component Connection Diagram (TS2100/SR–405HC–L1550)



Component connection diagram